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1 BRINGING THE LAB TO THE LAND WITH REAL-TIME SOIL ANALYSIS

Pune-based Proximal Soilsens Technologies bridges the gap between lab testing and the field with proximal sensors for real-time soil analysis.



NutriSens compact soil testing machine



TerraPort-IoT portable moisture meter

Proximal Soilsens Technologies Private Limited, founded by Rajul Patkar in June 2017, has introduced two innovations: NutriSens and TerraPort. NutriSens is a device designed to detect soil nutrients within five minutes, offering instant recommendations for farmers. TerraPort, a compact 100-gram portable device equipped with multiple sensors, enables farmers to traverse their fields, gather real-time data, and upload it via mobile for immediate insights.

"With just 30 minutes of training, anyone—including farmers, women, and students—can operate both devices. NutriSens is a lightweight, 50-gram, ultra-low-power device that runs on mobile phone power and has been validated by Tamil Nadu Agricultural University, achieving an impressive 80-90%

accuracy compared to lab results," explains Rajul.

Previously, the startup developed TerraStation, an IoT-enabled soil moisture sensor that continuously transmits data to the cloud for remote irrigation management. However, it was a fixed installation, collecting data only from a specific field. Recognising the need for a more mobile solution, particularly for small farmers, the company introduced TerraPort. To monitor soil moisture profiles, farmers insert the device into the soil, enabling real-time data collection across multiple farms simultaneously. It functions without requiring electricity or batteries.

TerraPort is priced at ₹20,000. So far, it has been introduced primarily to raise awareness among farmers, with only 70 to 80 units sold.

Rajul shares that NutriSens is

a newly launched product, with approximately 180 to 200 units sold worldwide over the past two years. This device operates without a battery or charging. Once connected to a phone, it draws power directly from it. Farmers can instantly connect, test, and receive an automated report on their phone, providing comprehensive insights into soil parameters, nutrient interpretation, and related data.

NutriSens functions similarly to a glucometer, simplifying the soil testing process so that farmers can conduct tests directly in the field. It is priced at ₹40,000 plus taxes, with per-parameter testing costing ₹40 to ₹50, as charged by the facility provider. Each parameter refers to individual nutrients, including pH, EC, nitrate, phosphate, potassium, and organic carbon.

"Additionally, we offer two mobile applications available in the Google Play Store," adds Rajul. The two mobile applications are the SoilSens FARMER App, designed specifically for farmers, and the SoilSens AGRIWISE App, tailored for agricultural experts, consultants, and service providers. While the FARMER App assists farmers in managing soil health and optimising crop productivity, the AGRIWISE App caters to professionals and organisations offering agricultural